GROUP 9 · EC8208 · SOFTWARE ARCHITECTURE

REOAGMS

Renewable Energy Operations, Asset and Grid Management System

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02 · ARCHITECTURE

One gateway, six independent services

React console · nginx · live updates over WebSocket

API Gateway · JWT validation · role policy

Identity	Asset	Monitoring	Maintenance	Notification	Analytics
PostgreSQL	PostgreSQL	PostgreSQL	PostgreSQL	PostgreSQL	PostgreSQL

RabbitMQ events · Redis cache · Prometheus and Grafana

LIVE Console: sign in, Control room

The registry every service relies on

Facility
5 types

›

Asset
7 types

›

Sensor
9 metrics

5 · 11 · 32

facilities · assets · sensors

Server-side

search and paging

Validated

every form

LIVE Assets page

04 · MONITORING SERVICE

From a sensor reading to an instant alert

Reading
every few seconds

›

Rule check
threshold

›

Alert
one per asset

›

RabbitMQ
event

16,000+
readings stored

7
alert rules

10 s
Redis cached dashboard

LIVE Live monitoring page (tiles and dashed threshold)

05 · MAINTENANCE SERVICE

Faults become tracked work automatically

Alert event
high or critical

›

Work order
created

›

Technician
assigned

›

Completed
in history

0 clicks
to open a corrective order

4
technicians

1
active order per asset

LIVE Maintenance board

Every alert delivered and recorded

Queue
durable

›

Consumer
schema check

›

Channels
email · push

›

History
stored

Instant
in-app push

3
automatic retries

100%
deliveries recorded

LIVE Notifications page

Trustworthy energy figures and reports

Source APIs — REST only › **Calculation** — power to kWh › **Reports** — stored › **CSV** — export

4 domain models

Dashboards
Energy Reports
Performance reports
KPIs

Dedicated PostgreSQL database

Pure Calculation engine

KPIs with targets

Five roles, enforced at the gateway

Administrator	Everything, users and audit log
Operations Manager	Assets, rules, work orders, reports
Plant Operator	Monitor and handle alerts
Maintenance Engineer	Assign and complete work orders
Executive	Read-only dashboards and reports

LIVE executive sign-in, read-only

09 · OPERATIONS AND QUALITY

Observable, tested, one command to run

60
test methods

7
services monitored

19
containers

1
command to start

LIVE Grafana platform health

Thank you

Questions and discussion

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